

Generative AI and Foundation Models in De Novo Protein Design: Regulatory Hurdles, Structural Validation, and Clinical Translation Pathways

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1 Introduction

Generative AI and foundation models have opened expansive, design-space horizons for de novo protein engineering, offering unprecedented potential to tailor sequence, structure, and function. Realizing this potential in the clinic, however, demands more than technical novelty: it requires an integrated regulatory science framework that spanning discovery, validation, and translational milestones across major jurisdictions. This report synthesizes how AI-driven design interacts with regulatory expectations in the FDA, EMA, and PMDA, and outlines the concrete pathways, evidence packages, and governance structures needed to de-risk de novo proteins from bench to bedside.

Central to this framework are five interlocking threads. First, phased, harmonized regulatory pathways that specify submissions, decision milestones, and go/no-go gates, with explicit attention to design provenance, data provenance governance, model versioning, and change control. Second, rigorous structural validation that couples AI-predicted models with experimental maps (for example, cryo-EM/X-ray data) and reports per-residue confidence (analogous to pLDDT), model-data fit metrics (FSC, map-to-model CC), and validation dashboards suitable for regulator review, including uncertainty indicators and traceable design rationales. Third, comprehensive evidence packages for GLP/GMP-compliant data generation, including appropriate toxicology, safety pharmacology, immunogenicity assessments, species bridging concepts, and the role of in silico risk predictors in risk assessment. Fourth, systematic GMP/GLP implications for the AI-driven design-to-manufacturing pipeline, addressing software as a medical device considerations, data lineage and cybersecurity, quality systems, supplier controls, and cross-border audits. Fifth, explicit treatment of regulatory gaps and cross-market alignment—data sharing, harmonized submission modules, upfront adoption of standards (ICH/ISO, NIST/ISO mappings, GAIP principles), and governance mechanisms enabling agile, parallel engagement across FDA, EMA, and PMDA.

Together, these strands frame a regulatory-ready, translational blueprint for AI-designed de novo proteins. [1]

2 Regulatory-ready blueprint for AI-driven de novo protein design

- 2.1** The regulator-ready blueprint presented here translates AI-driven de novo protein design into an integrated regulatory pathway that spans discovery, validation, translational milestones, and market authorization. It aligns the five interlocking threads—phased, harmonized pathways; rigorous structural validation with AI-predicted models anchored to experimental maps; GLP/GMP-compliant data gen and immunogenicity/toxicology considerations; GMP/GLP implications for the AI-enabled design-to-manufacturing pipeline including software as a medical device (SaMD) considerations; and explicit governance for cross-market alignment and data sharing—into a practical, risk-based framework suitable for FDA, EMA, and PMDA interactions. The discussion synthesizes current regulatory science and methodological precedent, drawing on peer-reviewed regulator-facing work, foundational de novo structure modeling, and AI governance concepts to outline evidence packages, decision gates, and governance structures that regulators can reasonably expect to see in credible filings.

[1]

2.2 Overarching design and submission philosophy

The path from discovery to potential marketing authorization should be staged through harmonized gates with explicit design provenance and change-control mechanisms, model versioning, and continuous regulator-facing documentation. This phased approach aligns with cross-market notions of regulator alignment and good-practice AI governance, as discussed in peer-reviewed analyses of AI in drug development and regulatory science [1]. These works emphasize the need for shared expectations about data provenance, decision gates, and transparency in AI-enabled submissions [1]. Additionally, foundational operational framing for de novo modeling—in particular, per-residue confidence indicators and model-data fit metrics in structure validation—has been advanced by Cryo2Struct, which demonstrates regulator-relevant validation frameworks for AI-assisted structural models derived from cryo-EM data [2]; and by subsequent PMC-hosted discussions and preprints that map onto standard structural biology validation practice [4]. The combination supports a phased framework with explicit evidence packages and dashboards for regulators [2].

2.2.1 Phase 0/0+ to Phase 1 (Discovery through IND/CTA readiness)

Milestones:

1. Milestone 0: Initial design provenance and governance setup; standard operating procedures (SOPs) for model training, data curation, versioning, and traceability; early alignment with FDA/EMA/PMDA through a formal scientific advice or parallel consultation plan where available [?].
2. Milestone 1 (0–12 months): Submission readiness assessment and go/no-go decision to enter preclinical GLP data generation; iterative design cycles with regulator-facing design rationales; approval to begin GLP/GMP studies subject to internal risk assessment and parallel scientific advice tracks [1].
3. Milestone 2 (12–24 months): IND/CTA enabling activities or PMDA equivalent initiation; validated AI-driven design with GLP-toxicology and safety pharmacology studies underway; concurrent establishment of GMP/QA readiness for manufacturing of AI-informed biologics or protein constructs [5].

4. Milestone 3 (24–36 months): Early-phase GLP tox/toxPK and bridging studies; regulator-facing data packages including design provenance, model version history, and uncertainty indicators (pLDDT-like scores, model confidence) in submissions; GMP manufacturing process validation in train with AI-enabled design inputs and traceability artifacts [2].
5. Milestone 4 (36–60 months): Readiness for MAA/PA/approval filings or PMDA final approvals; parallel market readiness with harmonized submission modules and cross-border data sharing approaches; post-approval pharmacovigilance scaffolding (if applicable) and continued model governance after market entry [1].

2.2.2 Phase 1/2 through Phase 3 (clinical development to potential marketing authorization)

Milestones:

1. IND/CTA/MAA decision gates: Interim go/no-go points tied to design provenance sufficiency, GLP/GMP data integrity, and regulator-graded validated evidence dashboards that integrate AI design rationales with experimental validation and safety data [?].
2. Deliverables: A regulator-facing dossier structure that includes (i) algorithm identity, training datasets, version history, audit trails, and change-control records; (ii) per-residue confidence metrics and map-to-model validations; (iii) GLP/GMP-compliant nonclinical data packages (toxicology, safety pharmacology, immunogenicity assessments, bridging strategies); (iv) software/SaMD governance and data lineage artifacts; (v) cross-market harmonization plans with shared submission modules.

2.3 Evidence packages and design provenance governance (what regulators expect to see)

A credible program presents design provenance for AI-driven de novo proteins, including a transparent chain of custody for design decisions, data used for training and testing, and a reproducible model versioning framework. The governance goal is to ensure regulators can reproduce and audit the rationale for design choices and any subsequent design iterations that affect product safety or efficacy. Foundational guidance supports this framing: AI-related regulatory submissions require clear design provenance, data provenance governance, and model-versioning control, as discussed in peer-reviewed regulatory science analyses [1], while AI-enabled structural validation frameworks emphasize per-residue confidence metrics and robust validation dashboards that regulators can examine directly within a submission [?]. In practice, dossiers should organize evidence around the following axes:

- Design provenance: algorithm identity, training/testing datasets, data curation steps, preprocessing, feature representations, and comprehensive version history; changelogs with explicit release notes and traceability to design rationales (why a given sequence was selected, what constraints guided changes) [1].
- Data provenance governance: audit trails for all data inputs, transformations, and storage; data lineage diagrams that show how each design parameter maps to data sources; compliance with data integrity principles and cross-border data handling considerations; and documentation of data access controls and cybersecurity measures [?].
- Model versioning and change control: SOPs governing model training, re-training, evaluation, and release; containerization and artifact repositories; verifiable release notes and roll-back capabilities; traceability from training seeds to final design outputs [?].
- Validation metrics and uncertainty indicators: per-residue confidence proxies (analogous to pLDDT), global model confidence, FSC and real-space CC for map-to-model fit, RMSD/TM-score metrics against reference experimental maps or structures; documentation of calibration approaches and uncertainty quantification; dashboards summarizing uncertainty alongside point estimates [?]; [?].

- Experimental validation dashboards and regulator-facing summaries: per-residue fit indicators, model–data congruence metrics, map quality indicators (resolution, completeness), and traceable design rationales linking experimental outcomes to design iterations [2].

2.4 Data generation, preclinical data expectations, and translational bridging

GLP-compliant studies are essential to establish safety and pharmacology baselines for AI-designed de novo proteins. The literature and regulatory discourse emphasize that GLP data generation should be complemented by careful bridging strategies and a risk-based approach to immunogenicity assessment:

- Nonclinical safety pharmacology and toxicology: Repeat-dose toxicology, toxicokinetics, biodistribution endpoints, and safety pharmacology studies tailored to the mechanistic features of de novo proteins. These studies should be designed with bridging concepts to anticipate human relevance, incorporating species selection justification and bridging strategies for immunogenicity and clearance patterns [5].
- Immunogenicity assessments: Given de novo sequences, explicit immunogenicity risk characterization is required; immunogenicity endpoints should be embedded in nonclinical plans and used to refine AI design loops where risk signals trigger design iterations [?].
- Bridging concepts and species selection: A strategy anchored in mechanistic modeling (PBPK/QSP) supports extrapolation from animal models to humans, enabling risk-based estimation of effective and safe dose ranges and exposure margins; this is supported by PBPK/QSP frameworks that emphasize bridging concepts and calibration against human-relevant endpoints [?].
- In silico risk predictors in risk assessment: In silico predictions can inform risk prioritization and study design, but regulators will expect them to be presented with traceable links to experimental verification and model governance artifacts [?].
- Immunogenicity-specific preclinical bridging: Because de novo proteins may present novel immunogenic features, immunogenicity assessments should be integrated into bridging strategies, with a plan for communicating residual uncertainty and risk in the submission package [?].

2.5 GMP/GLP implications for the AI-driven pipeline; SaMD considerations and data integrity

A cohesive program must address software- and data-centric quality systems throughout discovery, design, and manufacturing:

- GLP/GMP data lineage, cybersecurity, and data integrity: Data lineage and traceability must be maintained across AI model development, experimental results, and data used in manufacturing decisions; cybersecurity controls for data and model artifacts are essential, including access controls and audit trails, to satisfy cross-border data transfer and regulatory audits [?].
- Software as a Medical Device (SaMD) considerations: AI-enabled design tools used in the clinical development or manufacturing decision space may constitute SaMD or SaMD-like components. Regulators will expect defined risk management, software verification/validation, cybersecurity, maintenance procedures, and post-market monitoring as appropriate [?].
- Quality systems and supplier controls: Documentation of supplier qualifications for data sources and for AI tooling; change controls for AI inputs that impact manufacturing controls; process validation strategies that tie AI outputs to manufacturing specifications [?].
- Data lineage, traceability, and cross-border audits: A robust governance framework should document data origin, transformations, model versions, and manufacturing change events with auditable trails to enable inspections and data-sharing across jurisdictions [?].

2.6 Cross-market harmonization; data sharing; and governance mechanisms enabling agility

The regulatory landscape is evolving toward cross-border alignment and harmonized submission modules, with early emphasis on joint principles and agility in engagement with FDA, EMA, and PMDA. Peer-reviewed analyses underscore the value of cross-market alignment and standardization efforts as a basis for smoother global submissions; pilots and mutual recognition concepts are commonly discussed as actionable pathways to reduce rework and duplicative effort [?]. A practical plan would include:

- Early joint scientific advice and parallel discussions among agencies to define common data dumps, dossiers structures, and validation dashboards that satisfy all three regulators [?].
- Adoption of harmonized submission modules and standards (ICH/ISO mappings; GAIP principles) to streamline cross-border data sharing and auditability; such standards provide a pragmatic scaffold for multi-jurisdictional reviews while allowing for jurisdiction-specific tailoring where necessary [?].
- Pilot programs to test cross-border data sharing and submission workflows, with explicit metrics for timeliness, rework, and decision gate alignment; governance bodies should monitor milestones and publish learnings to accelerate alignment [?].

2.7 Gaps in current regulatory guidance and actionable pathways to address them

The current landscape lacks finalized, AI-specific regulatory guidance for de novo proteins. However, a convergent view across credible sources points to several actionable pathways:

- Standards adoption: Immediately anchoring on ICH/ISO frameworks for data provenance, risk management, and quality by design; paired with NIST AI RMF and ISO 42001 to operationalize governance artifacts, while maintaining flexibility for evolving guidance [?].
- Cross-border audits and data sharing: Establishing collaborative pilots to test harmonized data sharing protocols, with shared evidence-pack templates and standardized dashboards to support parallel reviews, subject to cyber and IP protections [?].
- Harmonized submission modules: Developing cross-market submission templates with standardized sections for design provenance, versioning, and uncertainty quantification so regulators can readily assess AI-driven rationale alongside experimental results [?].
- Governance mechanisms enabling agility: A governance framework that integrates red-teaming, model risk management, and traceability artifacts (model cards, data licenses, risk registers) aligned to ISO/NIST guidance would support rapid iterations without compromising regulatory confidence [?].
- Audits and inspection readiness: Proactive preparation of inspection-readiness artifacts—data lineage traces, access-control logs, version histories, and regulatory dashboards—so that regulators can trace decisions from AI inputs to clinical outcomes [?].

2.8 Practical steps a sponsor should take early to harmonize across markets

- Engage early with regulators through parallel scientific advice and similar pre-submission dialogues to establish scope and expectations for AI-derived design rationale, model governance, and data lineage artifacts [?].
- Build a unified governance framework anchored to NIST RMF and ISO 42001, with explicit mapping to data provenance and model-card artifacts that regulators can cross-compare across jurisdictions [?].
- Establish a shared evidence-pack architecture: a regulator-facing dossier blueprint containing consistent sections for design provenance, versioning, uncertainty quantification (pLDDT-like proxies; FSC/real-space CC), GLP/GMP plans, and cross-market data sharing concepts; ensure this blueprint remains flexible for jurisdiction-specific tailoring [?].

- Implement parallel, cross-discipline governance bodies that include AI ethics/safety leads, regulatory affairs, quality systems, and clinical development to ensure that design iterations, risk assessments, and data handling are integrated from the outset [?].
- Institute a robust data lineage and change-management discipline: maintain traceability from training data through model versions to final design outputs, with explicit release notes and audit trails that regulators can review as part of the submission package [?].
- Prepare regulator-facing dashboards and evidence packages that can be individually adapted for FDA, EMA, PMDA while maintaining core harmonized elements, thereby reducing rework when a given market requires additional justification or alternative thresholds [?].

3 Regulatory-submission architecture: evidence packages, dashboards, and data governance artifacts

3.1 Evidence package structure

The evidence package should interlock design provenance with validated structural data, nonclinical data, and manufacturing governance artifacts. The key components include:

- Design provenance dossier: algorithm identity, datasets, data curation steps, version history, release notes, and traceability to specific design decisions [?].
- Data provenance and auditability: complete data lineage diagrams, access controls, transformational logs, and data integrity attestations; cross-border data-sharing considerations and compliance with data privacy and security standards [?].
- Model versioning and change control: SOPs for training, evaluation, and release; fully auditable traceability from seeds to final design outputs, with rollback capabilities for safe reversion if needed [?].
- Validation dashboards for regulators: per-residue confidence proxies (pLDDT-like metrics), model-data fit metrics (FSC, map-to-model CC), calibration/uncertainty quantification, and narrative design rationales linking design choices to experimental outcomes [?].
- Data generation and nonclinical evidence: GLP-compliant study designs, bridging strategies, and immunogenicity risk assessments; endpoints, durations, species rationale, and bridging strategies to humans; plan for integrating in silico predictions in risk assessment and decision gates [?].
- GMP/GLP and manufacturing artifacts: quality systems, software validation artifacts, supplier controls, cybersecurity, process validation plans, and change-management records for AI-driven design influence on manufacturing specifications [?].
- Cross-market harmonization artifacts: joint submission module definitions, pilot programs, data-sharing agreements, and alignment with ICH/ISO mappings to support consistent regulatory reviews across FDA/EMA/PMDA [?].

3.2 Uncertainty communication and validation dashboards

- Per-residue confidence (pLDDT-like proxies) and model confidence
- FSC curves and real-space CC across resolution shells to demonstrate map-to-model fit
- Calibration curves and uncertainty bounds for structural predictions
- Regulator-facing visualizations with clear escalation paths when discordance persists, including traceable design rationales and the path to iteration or freeze points
- Documentation of data provenance and model versioning; explicit trails from training data through to final design decisions

4 Data generation, preclinical data and translational bridging

4.1 GLP-compliant nonclinical data package

- Repeat-dose toxicology, toxicokinetics, biodistribution, and immunogenicity assessments tailored to de novo designs
- Bridging concepts: preclinical-to-clinical extrapolation supported by PBPK/QSP frameworks that inform dose selection and safety margins; alignment with MDPI Tier 1–4 PBPK/QSP calibration frameworks to enable translational bridging and regulatory justification [?].
- Immunogenicity risk management: immunogenicity assessments should be integrated into non-clinical plans with explicit decision gates to drive AI redesign if risk signals exceed predefined thresholds [?].
- Species selection and bridging strategy: bridging plans designed to translate exposure, immunogenicity, and safety signals to humans; PBPK/QSP-guided bridging informs early clinical dose ranges [?].

5 GMP/GLP considerations; SaMD and data integrity

5.1 Data lineage and cybersecurity

End-to-end traceability for AI inputs, model updates, and resulting design decisions; robust data lineage diagrams and audit trails to satisfy inspections in cross-border contexts [?].

5.2 SaMD considerations

Where AI design tools influence clinical development decisions, SaMD governance frameworks should be in place, including risk management, software verification/validation, cybersecurity, maintenance procedures, and post-market monitoring as appropriate [?].

5.3 Quality systems and supplier controls

Documentation of supplier qualifications for data sources and for AI tooling; change controls for AI inputs that impact manufacturing controls; process validation strategies that tie AI outputs to manufacturing specifications [?].

5.4 Cybersecurity, data integrity, and audit readiness

Cybersecurity controls for data and model artifacts; access controls with auditable logs; cross-border audit readiness through standardized artifacts and dashboards [?].

6 Gaps, alignment, and actionable pathways to address them

6.1 Regulatory guidance gaps

No finalized, AI-specific regulatory guidance for de novo proteins currently exists in many jurisdictions; however, a convergent set of best practices is emerging via cross-agency discussions and governance frameworks. The literature underscores the importance of cross-market alignment and joint guidance on design provenance, data governance, and model risk management [?].

6.2 Actionable pathways

- Standards adoption: anchor on ICH/ISO frameworks for data provenance, risk management, and quality by design; align with NIST AI RMF and ISO 42001 to operationalize governance artifacts; maintain flexibility for evolving guidance [?].

- Cross-border audits and data sharing: establish collaborative pilots to test harmonized data sharing protocols, with shared evidence-pack templates and standardized dashboards to support parallel reviews, subject to cyber and IP protections [?].
- Harmonized submission modules: develop cross-market submission templates with standardized sections for design provenance, versioning, and uncertainty quantification to facilitate regulator review [?].
- Governance mechanisms enabling agility: a governance framework that integrates red-teaming, model risk management, and traceability artifacts (model cards, data licenses, risk registers) aligned to ISO/NIST guidance would support rapid iterations without compromising regulatory confidence [?].
- Audits and inspection readiness: prepare inspection-ready artifacts—data lineage traces, access-control logs, version histories, and regulator dashboards—to enable traceability from AI inputs to clinical outcomes [?].

7 Milestone-based governance and practical actions for sustained agility

7.1 Governance architecture

A cross-functional governance body should oversee data provenance, model versioning, safety and risk management, and cross-market submission readiness; integrate red-teaming and risk assessment artifacts into the regulatory dossier.

7.2 Key metrics and KPIs

Timeliness of go/no-go gates; rate of iterations per design cycle; auditability scores for data lineage and model governance artifacts; proportion of cross-market alignment milestones achieved in planned timelines; regulator satisfaction with dashboards and evidence packs.

7.3 Instrumentation

Provenance schemas, model-card templates, risk registers, test plans, and audit reports aligned to ISO/NIST expectations; dashboards to track artifacts and governance metrics; version-control and containerization for reproducibility.

8 Regulator-facing synthesis and conclusion

The pathway outlined here situates AI-designed de novo proteins within a structured, regulator-facing framework that emphasizes design provenance, robust validation dashboards, GLP/GMP data packages, and cross-market harmonization. It integrates Cryo2Struct’s explicit emphasis on residue-level confidence and map-to-model validation with established PBPK/QSP bridging concepts and de novo modeling foundations from Nat Methods 2015. It acknowledges current regulatory gaps while offering practical, implementable pathways through standardized data provenance, model governance, and regulator-ready evidence packages. As AI-driven de novo protein design matures, the blueprint above provides a roadmap to harmonize design provenance, experimental validation, and manufacturing governance in a way that regulators can audit, reproduce, and cross-check across FDA, EMA, and PMDA review processes. The emphasis on per-residue uncertainty indicators, explicit model versioning, and cross-border data sharing is designed to reduce rework and accelerate decision gates, while maintaining rigorous risk management and traceability that regulators increasingly demand in next-gen biologics programs. In sum, a harmonized, phased regulatory pathway—bolstered by rigorous governance, transparent design provenance, and regulator-facing evidence dashboards—offers a credible, scalable framework to bring AI-designed de novo proteins from bench to bedside across major markets [?].

References

- [1] AI in Regulatory Submissions and Regulatory Science: Cross-Jurisdictional Considerations. <https://pmc.ncbi.nlm.nih.gov/articles/PMC12675821>
- [2] Cryo2Struct: De Novo Atomic Modeling for Cryo-EM Density Maps. <https://pmc.ncbi.nlm.nih.gov/articles/PMC11217428>
- [3] Cryo2Struct: De Novo Atomic Modeling for Cryo-EM Density Maps (Code Repository). <https://github.com/jianlin-cheng/Cryo2Struct>
- [4] Wang, N. De Novo Protein Structure Modeling: Foundational Frameworks. Nat Methods 2015. https://www.bakerlab.org/wp-content/uploads/2015/12/Wang_NatMethods_2015.pdf
- [5] PBPK modeling in drug development and translational risk assessment. PubMed. <https://pubmed.ncbi.nlm.nih.gov/42196328>
- [6] PBPK/QSP bridging framework (MDPI): Tier 1–4 validation and calibration. <https://www.mdpi.com/1422-0067/27/10/4349>
- [7] Cross-market AI governance and regulatory alignment discussions. SSRN abstract 6740438. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6740438